

GIRISH CHANDAR G

East Lansing, Michigan

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OBJECTIVE

Motivated Ph.D. student interested in research internships in Monocular depth estimation, 3D reconstruction, 3D generation and novel-view synthesis. I have relevant academic knowledge and research experience in Diffusion models, NeRFs, and 3D Gaussian Splatting.

EDUCATION

Michigan State University, MI, USA <i>Ph.D. Computer Science, Advisor: Dr. Xiaoming Liu</i>	Jan 2024 – Present GPA : 4/4
University of Michigan, Ann Arbor, MI, USA <i>M.S. Electrical and Computer Engineering</i>	Aug 2021 – Apr 2023 GPA : 4/4
Indian Institute of Technology Gandhinagar, Gandhinagar, India <i>B.Tech. Electrical Engineering (minor Computer Science)</i>	July 2016 – Aug 2020 GPA : 8.98/10

COURSEWORK / SKILLS

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|---------------------------------------|---|---|------------------------------|----------------------------------|
| • Foundations for Computer Vision (A) | • Matrix Methods for Machine Learning and Signal Processing(A+) | • 3D Computer Vision (A) | • PyTorch, Tensorflow, MXNet | • Numpy, OpenCV, Sklearn, Pandas |
| • Machine Learning (A) | | • Deep Learning for Computer Vision (A) | • MATLAB, LabVIEW | • C++ |

PUBLICATIONS

DiVER: Disentangled Visual Embeddings for Robot Manipulation <i>Girish Chandar G, Yuliang Guo, Xiaoming Liu</i>	In Progress
DepthAgent: Towards Better Universal Depth Estimation via Sample-wise Expert Selection 📄 <i>Jie Zhu, Girish Chandar G, Xiaoming Liu</i>	In Review, NeurIPS 2026
UniDAC: Universal Metric Depth Estimation for Any Camera 📄 <i>Girish Chandar G, Yuliang Guo, Liu Ren, Xiaoming Liu</i>	CVPR 2026
Unleashing the Power of Chain-of-Prediction for Monocular 3D Object Detection 📄 <i>Zhihao Zhang, Abhinav Kumar, Girish Chandar G, Xiaoming Liu</i>	CVPR 2026
RePLAY: Remove Projective LiDAR Depthmap Artifacts via Exploiting Epipolar Geometry 📄 <i>Shengjie Zhu*, Girish Chandar G*, Abhinav Kumar, Xiaoming Liu</i>	ECCV 2024

POSITION

Research Assistant Dr. Xiaoming Liu, Michigan State University 📄	June 2023 - Present
Graduate Student Research Assistant Dr. Stella Yu, University of Michigan-Ann Arbor 📄	Jan 2023 – Apr 2023
Research Intern Dr. Yang Zheng, NVIDIA 📄	May 2022 – August 2022

INTERNSHIPS

Long-Range Lane Detection for Autonomous Driving Stack AV PyTorch	May 2026 - Present
Stereo Hazard Detection NVIDIA PyTorch - Implemented end-to-end deep learning model using custom UNet as the backbone for feature map extraction. - Outperformed baseline by achieving zero false positives.	May 2022 – Aug 2022
Auto Shape Detection in Machine Vision 📄 Zentron Labs Python (Numpy, OpenCV) - Implemented Arc Detection algorithm that gives accuracies of 100% on simulated data and 80% on real data. - Improved Line and Circle Detection accuracies from 65% to 90%	Oct 2020 – Aug 2021
Optimization based Inverse Rendering 📄 The University of Texas at Dallas PyTorch - Implemented algorithm for 3D face reconstruction from 2D images. 3D Morphable Model (3DMM) used as apriori mesh for efficient inverse rendering.	May 2019 – July 2019

PROJECTS

SAR-NeRF PyTorch Research focused on modifying NeRF for 3D reconstruction of complex-valued radar data.	Apr 2023
Small NeRF 📄 PyTorch - Implemented a modified version of NeRF to reduce training time and computational cost. - Experimented with multiple architectures to determine the best approximation of the original NeRF.	Apr 2022
Co-Tuning for Transfer Learning on TACO Dataset 📄 PyTorch - Implemented and verified the novel transfer algorithm proposed in "Co-tuning for Transfer Learning". First team to implement co-tuning on TACO (Trash Annotations in Context) dataset.	Dec 2021